

Andrew Koja

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Education

Michigan State University

Aug 2024 – May 2028

Bachelor of Science in Computer Science

East Lansing, MI

- **GPA:** 3.53 Cumulative, 3.83 Major
- **Relevant Coursework:** Algorithms & Data Structures (*In Progress*), Computer Organization & Architecture (*In Progress*), Programming II (C++), Discrete Structures, Programming I (Python), Calculus I–III

Experience

Apostles4Jesus

June 2023 – Present

Full-Stack Software Developer

Macomb, MI

- Designed and deployed a bilingual (English/Arabic) podcast platform for an organization with 15,000+ social media followers using HTML/CSS, JS, and Netlify
- Mitigated OWASP Top 10 risks with CSP, rate limiting, input sanitization, and reCAPTCHA via serverless Netlify Functions
- Implemented WebP conversion and lazy loading to cut homepage payload 10x (3.3MB to 336KB), improving load times for mobile users on slow connections

Projects

LightAPI | *React, FastAPI, PyTorch, SQLite, WebSockets, Docker*

lightapi.dev

- Engineered a searchable directory of 1,500+ public APIs where any entry becomes a live monitor in one click, processing 28,800+ daily latency readings under 100ms
- Developed per-endpoint LSTM models in PyTorch to detect latency anomalies, plus a root cause analysis engine linking regressions to GitHub deployments
- Shipped a pip-installable Python SDK featuring a non-blocking `@monitor` decorator, backed by a containerized FastAPI service

CrimeScope | *React, Vite, Leaflet, Zustand, Recharts, SODA / ArcGIS APIs*

crime-scope.vercel.app

- Normalized data from 17 U.S. city APIs, enabling client-side filtering of 50,000+ records without a backend
- Rendered an interactive visualization dashboard featuring pin, cluster, and heatmap mapping modes
- Implemented shareable URL state, stale-data detection, and city-aware date handling across 17 open-data portals

FootyLive | *Python, FastAPI, React, TypeScript, SQLite, WebSockets, Redis*

FootyLive

- Architected a real-time soccer dashboard covering 12 competitions, delivering live scores every 30 seconds via WebSockets with automatic polling fallback
- Developed broadcast-style knockout brackets for the Champions League, Euros, and World Cup with two-leg aggregation and penalty-shootout resolution
- Maintained uptime through background synchronization, in-memory caching, and season archiving that serves finished seasons without external API calls

FaceRead | *React, Vite, face-api.js, MediaPipe, Web Audio API, WebGL*

face-read.vercel.app

- Deployed face-api.js fully client-side via WebGL to run 3 neural networks in parallel at 15–30 FPS with zero server lag
- Built a browser-based ASL fingerspelling classifier using MediaPipe Hands keypoints supporting all 26 letters
- Reduced inference latency from 500ms to real-time by migrating the pipeline from a Flask backend to client-side WebGL

Technical Skills

Languages & Frameworks: Python, JavaScript, TypeScript, Java, C++, HTML/CSS, SQL, Flask, FastAPI, React, Vite, Tailwind CSS, Node.js

Tools & DevOps: Git/GitHub, GitHub Actions (CI/CD), Docker, pytest, SQLite, Redis, PyTorch, WebSockets, SQLAlchemy, Vercel, Render

Concepts: Machine Learning, Real-Time Systems, Anomaly Detection, REST APIs, Data Visualization, Server-Side Caching